Mind the Touch Gap: Real-to-Sim Contact Alignment for Dexterous Pen Spinning

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Abstract: Pen spinning is fast and contact-rich, making simulation transfer difficult. PenSpin [1] fine-tuned a proprioceptive student on successful hardware replays. We ask whether a small replay budget can instead align the contact observations used during simulator training. Using Isaac Gym [2] and Project SuperDex [3], we propose to learn a source-to-target contact mapper, re-distill a contact-aware student, and test it against proprio-only, raw-contact, randomized-contact, calibrated-contact, and target-fine-tuning baselines. This sim-to-sim study tests the value of contact-observation alignment without claiming to correct dynamics or establish hardware transfer.

Keywords: Dexterous manipulation, contact alignment, sim-to-sim transfer

1 Introduction

Simulation-based learning has enabled dexterous in-hand skills [4, 5, 6], but rapid contact changes still impede transfer. PenSpin [1] trained a privileged oracle and a proprio-only student; the student did not transfer reliably to hardware.

PenSpin then fine-tuned the student on successful open-loop hardware replays. We instead ask whether the same limited target experience can improve the *contact observations* used for training, so a student can learn from target-like touch cues in the source simulator.

Prior work adapts simulator parameters from target experience [7, 8, 9] or calibrates tactile simulation through replay [10]. Our intervention aligns only the policy's contact input; the source collision solver and dynamics remain unchanged. Because PenSpin's student had no contact input, we add a contact-aware student and retain the original as a baseline.

We use Isaac Gym as the source and SuperDex [3] as the target. SuperDex provides contact labels and controlled resets, allowing matched target tests and friction/compliance changes. This is a controlled proxy rather than a hardware result.

We ask three questions:

RQ1: When does contact help or hurt transfer under friction and compliance shifts?
RQ2: How much can a small replay-trained contact mapper improve target performance?
RQ3: At equal replay attempts, how does alignment compare with target-policy fine-tuning?

Our plan retains PenSpin's oracle and proprio-only student, adds a student with binary link contact, and learns a contact mapping from paired replays. We then freeze the mapper, re-distill in the source simulator, and test in the target. The intuition is that target-like contact cues can improve learning even when the source physics cannot be corrected by observation alignment.

Course Project, 16-831 Introduction to Robot Learning, Fall 2026, CMU

2 Plan

Implementation and data collection. We will use PenSpin's Allegro hand, pen-like objects, action space, and PD control. Corresponding hand links in Isaac Gym ($\mathcal{S}$) and SuperDex ($\mathcal{T}$) will share a binary contact representation at the policy observation rate. We will match object geometry, control timing, initial states, and rotation/drop criteria. After training the privileged oracle and the two students, we will replay the same oracle actions in both simulators from matched initial states and record contact, joint and object states, and preceding actions. Every attempted replay supplies mapper labels, including failed attempts.

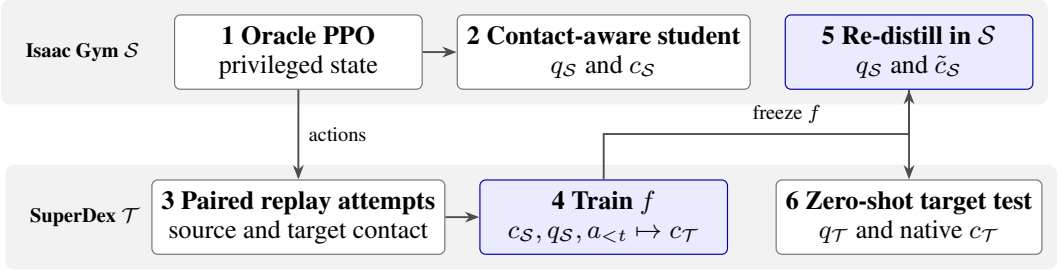


Figure 1: Six-stage proposal plan. Paired replay attempts train the contact mapper; source rollouts re-distill the student. SuperDex supplies native contact for evaluation. Target-policy fine-tuning is a separate baseline.

Contact alignment. A small temporal mapper $f_\phi(c_{\mathcal{S},t-H:t}, q_{\mathcal{S},t-H:t}, a_{t-H:t-1})$ predicts target link-contact probabilities from source contact c, proprioception q, and action history a available before action a_t. We train it with binary cross-entropy on paired replay contacts, choose per-link thresholds on validation replays, and supply the resulting binary contacts when re-distilling the student from the fixed oracle in $\mathcal{S}$. The mapper is frozen during re-distillation and is not run in $\mathcal{T}$; deployment uses native target contact.

Evaluation and baselines. We will compare our aligned-contact student with the original proprio-only student, a raw-contact student, contact noise/dropout/delay randomization, parametric threshold-and-delay calibration [10], and PenSpin-style student fine-tuning on successful target replays [1]. All policies will use the same held-out SuperDex test objects, initial states, and episodes. Methods using target data receive equal numbers of *attempted* replay episodes; we will also report how many successful demonstrations the fine-tuning baseline retained. This keeps the data budget comparison interpretable without treating target fine-tuning as zero-shot.

Measurements and diagnostics. At budgets of 10, 25, 50, and 100 attempted target replays, we will measure full rotations, time until drop, and success rate, along with held-out per-link contact precision, recall, F1, and onset timing error. We will report variation across evaluation seeds and vary target friction and compliance. Because time-indexed open-loop replays can drift into different physical states, we will measure joint/object-state divergence and group contact errors by divergence. As a separate diagnostic, privileged resets will create state-matched contact pairs; this control requires extra simulator access and is not a deployable baseline or a mathematical upper bound.

Schedule and risks. The largest engineering risk is porting the Allegro/pen environment to SuperDex. We begin with a short hand-control/contact replay smoke test before scaling data collection.

- **10/09–10/23:** Reproduce the PenSpin oracle and proprio-only student; implement and validate the shared link-level contact representation.
- **10/24–11/06:** Port the Allegro hand, pen, and PD controller to SuperDex; match control timing and collect paired replay attempts.

- **11/07–11/20:** Train the contact mapper, re-distill the contact-aware student, and obtain first zero-shot target results.
- **11/21–12/04:** Run equal-attempt baselines, data-budget sweeps, friction/compliance variations, and state-matched diagnostics.
- **12/05–12/11:** Prepare the final report and presentation.

Fallbacks. If full pen spinning is unstable in the port, we will first test a simpler in-hand rotation task [4]. If the SuperDex port stalls, we will use a target simulator with explicit link-level contact labels and state the change. If the target gap is too small, we will widen it through controlled friction and compliance changes. These outcomes would remain simulation evidence rather than hardware transfer.

Acknowledgments

ChatGPT assisted with editorial revision and LaTeX formatting.

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